Two spins coupled to an XXZ bulk

Reduced dynamics and quantum-state texture

Spin transport project

November 6, 2026

Abstract

We study the joint reduced dynamics of two spin-1/2 probes, each attached to one end of a finite XXZ bulk. The total Hamiltonian generates unitary evolution, and the initial probe–bulk state is a product. For Ising boundary coupling, four conditional bulk Hamiltonians give an exact time-dependent Schur, or generalized dephasing, channel. Its complete action is encoded in the output from the single probe preparation $|{+}{+}\rangle$. At every time where it is invertible, this channel has a canonical time-local master equation with at most three nonzero decoherence rates and diagonal Lindblad operators, all given in closed form by the coherence factors of the Schur multiplier. For exchange coupling, the available description gives the symmetry blocks rather than closed expressions for their coefficients. When the Hamiltonian conserves total magnetization and the bulk preparation commutes with bulk magnetization, the reduced superoperator and Choi matrix split into blocks of sizes $1, 4, 6, 4, 1$. Quantum-state texture is evaluated only on the output from $|{+}{+}\rangle$; exact relations to the Ising channel, its fidelities and its entanglement-generating ability follow from the Schur structure.

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1 Introduction

Two probes coupled to a common spin chain form an open quantum system whose reduced dynamics depends on the interaction at each contact. We consider Ising and exchange contacts separately, with no direct probe–probe interaction. The bulk starts in a fixed state, uncorrelated with the probe pair. Tracing out the bulk after unitary evolution defines a quantum channel on the pair.

Ising contacts conserve both probe z projections and produce generalized dephasing in the joint computational basis. Exchange contacts allow spin transfer between the probes and the bulk. Conservation of total magnetization then constrains the reduced channel when the bulk preparation respects the same rotation symmetry.

These two interactions lead to the following channel descriptions:

1. Ising coupling: an exact conditional-Hamiltonian reduction, an exact Schur channel, and reconstruction of that channel from the full output state for $|{+}{+}\rangle$.
2. Ising coupling, time-local description: the canonical form of the master equation, with its decoherence rates, Lindblad operators and Hamiltonian expressed through the coherence factors.
3. Exchange coupling: exact total-magnetization sectors and, for a magnetization-invariant bulk preparation, a phase-covariant channel with five coherence-order blocks. These constraints leave the block entries to be computed from the many-body evolution.
4. Texture: exact grand-sum, return-probability and response-operator identities. For Ising coupling these yield channel fidelities and other properties without additional input preparations.

Section 2 fixes the model, the reduced channel and the texture conventions shared by both interactions. The two couplings are then treated in separate parts: Part I contains the Ising channel (Section 3) and its canonical form (Section 4); Part II contains the exchange channel (Section 5).

The derivations use exact finite-system dynamics, without a weak-coupling or Markov approximation. The channel structure fixes which probe operators can mix; their time dependence depends on the bulk Hamiltonian and its initial state.

Notation. To avoid collisions between symbols, we use the following conventions throughout.

Symbol	Meaning
$\Gamma(t)$, $\Gamma_{ab}(t)$	Ising Schur matrix and its complex, dimensionless entries
$\eta_{ab}(t) = \dot{\Gamma}_{ab}/\Gamma_{ab}$	logarithmic derivative of a coherence factor
κ_{ab}, ω_{ab}	coherence damping rate and phase rate, $\eta_{ab} = -\kappa_{ab} + i\omega_{ab}$
$\mathsf{d} = S + iA$	3×3 decoherence matrix on the diagonal operator sector
$\gamma_k(t)$	canonical decoherence rates (eigenvalues of d)
F_μ, $G = (g^a_\mu)$	diagonal traceless operators and their eigenvalue matrix
$H_{\text{eff}} = \text{diag}(h_a)$	canonical (effective) probe Hamiltonian
$\xi_{\pm 2}$	exchange-channel multiplier of the one-dimensional sectors $q = \pm 2$

Configuration labels a, b, c, d run over $00, 01, 10, 11$. The index $\mu, \nu \in \{1, 2, 3\}$ labels F_1, F_2, F_3; k labels canonical modes. The couplings g_A, g_B are unrelated to the eigenvalues g^a_μ.

2 Model, conventions and the reduced channel

2.1 Geometry and Hamiltonians

Let $N \geq 2$ be the number of *bulk* spins. There are $N + 2$ spins in total, with physical order $A, 1, \ldots, N, B$. We use

$$S^\alpha = \sigma^\alpha/2, \qquad S^\pm = S^x \pm iS^y = (\sigma^x \pm i\sigma^y)/2, \qquad \hbar = 1. \tag{1}$$

The basis convention is $|0\rangle = |\uparrow\rangle$, $|1\rangle = |\downarrow\rangle$, and the probe basis is $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. For formulas involving a product of probe and bulk operators, we group the Hilbert space as $\mathcal{H}_{AB} \otimes \mathcal{H}_E$ by a fixed permutation of the physical site ordering.

The reference bulk and the two alternative interactions are

$$H_E = J \sum_{j=1}^{N-1} (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z), \tag{2}$$

$$V_{\mathrm{ZZ}} = g_A S_A^z S_1^z + g_B S_N^z S_B^z, \tag{3}$$

$$\begin{aligned} V_{\mathrm{ex}} &= g_A (S_A^x S_1^x + S_A^y S_1^y) + g_B (S_N^x S_B^x + S_N^y S_B^y) \\ &= \frac{g_A}{2} (S_A^+ S_1^- + S_A^- S_1^+) + \frac{g_B}{2} (S_N^+ S_B^- + S_N^- S_B^+). \end{aligned} \tag{4}$$

The second line of Eq. (4) follows from $S^+ S'^- + S^- S'^+ = 2(S^x S'^x + S^y S'^y)$ for commuting spins S and S'. Thus $H^{(r)} = H_E + V_r$, with $r = \mathrm{ZZ}$ or ex. Identity operators on the remaining spins are implicit. The parameters J, g_A, g_B have energy units and Δ is dimensionless. Equal couplings mean $g_A = g_B = g$. The exchange flip-flop matrix element at the left contact is $g_A/2$ in this convention.

Equation (2) has no applied fields. Longitudinal bulk or probe fields preserve the magnetization arguments below, with their phases included in the conditional dynamics. Transverse *bulk* fields also preserve the Ising Schur reduction, but generally invalidate the exchange magnetization reduction because they break conservation of total z magnetization.

2.2 Product preparation and exact channel

Fix any bulk density matrix ω_E, independent of the probe input. The initial state and reduced evolution are

$$\rho(0) = \rho_{AB}(0) \otimes \omega_E, \tag{5}$$

$$U_t = e^{-iH^{(r)}t}, \qquad \Lambda_t(X) = \mathrm{Tr}_E[U_t(X \otimes \omega_E)U_t^\dagger], \tag{6}$$

$$\rho_{AB}(t) = \Lambda_t(\rho_{AB}(0)). \tag{7}$$

The product assumption separates the pair from the bulk; it imposes no factorization between A and B for the definition of the channel. Only the texture protocol fixes their input to $|{+}{+}\rangle$. The bulk state ω_E may be pure or mixed, thermal or nonthermal, and need not commute with H_E or with the bulk magnetization unless a specific result states otherwise. Changing ω_E changes the channel.

To see complete positivity and trace preservation directly, write $\omega_E = \sum_\nu p_\nu |\phi_\nu\rangle\langle\phi_\nu|$ with $p_\nu \geq 0$, $\sum_\nu p_\nu = 1$, and choose any orthonormal bulk output basis $\{|r\rangle\}$. Inserting $I_E = \sum_r |r\rangle\langle r|$ into the partial trace,

$$\Lambda_t(X) = \sum_{r,\nu} p_\nu \, {}_E\langle r|U_t|\phi_\nu\rangle_E \, X \, {}_E\langle\phi_\nu|U_t^\dagger|r\rangle_E. \tag{8}$$

Each ${}_E\langle r|U_t|\phi_\nu\rangle_E$ is an operator on $\mathcal{H}_{AB}$. Hence

$$K_{r\nu}(t) = \sqrt{p_\nu} \, {}_E\langle r|U_t|\phi_\nu\rangle_E, \quad \Lambda_t(X) = \sum_{r,\nu} K_{r\nu} X K_{r\nu}^\dagger, \tag{9}$$

and

$$\sum_{r,\nu} K_{r\nu}^\dagger K_{r\nu} = \sum_\nu p_\nu \, {}_E\langle\phi_\nu|U_t^\dagger \Big(\sum_r |r\rangle\langle r|\Big) U_t|\phi_\nu\rangle_E = \sum_\nu p_\nu \, \langle\phi_\nu|\phi_\nu\rangle \, I_4 = I_4. \tag{10}$$

These are formal exact expressions in both regimes; evaluating them for an interacting bulk remains a many-body problem. Correlated probe–bulk preparations have reduced trajectories, but do not in general define the same CPTP map on arbitrary probe inputs.

2.3 Choi normalization

The *normalized* Choi state, with the output factor first, is

$$|\Phi_4\rangle = \frac{1}{2}\sum_{a=1}^{4}|a\rangle_{AB}|a\rangle_R, \qquad J_t = (\Lambda_t \otimes \mathrm{id}_R)(|\Phi_4\rangle\langle\Phi_4|). \tag{11}$$

Since $|\Phi_4\rangle\langle\Phi_4| = \frac{1}{4}\sum_{a,b} E_{ab}\otimes E_{ab}$ with $E_{ab} = |a\rangle\langle b|$,

$$J_t = \frac{1}{4}\sum_{a,b}\Lambda_t(E_{ab})\otimes E_{ab}. \tag{12}$$

It obeys $J_t \succeq 0$, $\mathrm{Tr}\, J_t = 1$ and $\mathrm{Tr}_{AB}\, J_t = I_R/4$. The corresponding unnormalized Choi matrix is $C_t = 4J_t$. The auxiliary system R represents a reference for the channel and is independent of the physical bulk.

2.4 Quantum-state texture of the $|{+}{+}\rangle$ preparation

Texture is defined relative to the probe computational basis. Its textureless state, and the sole probe preparation used for texture, is

$$|f_1\rangle = |+\rangle_A \otimes |+\rangle_B = \frac{1}{2}\sum_{a=1}^{4}|a\rangle, \qquad f_1 = |f_1\rangle\langle f_1|, \qquad (f_1)_{ab} = \frac{1}{4}. \tag{13}$$

Denote its full output state by $\rho_f(t) = \Lambda_t(f_1)$. For any probe state, the grand sum is real and satisfies

$$\Sigma(\rho) = \sum_{a,b}\rho_{ab} = 4\langle f_1|\rho|f_1\rangle, \qquad 0 \le \Sigma(\rho) \le 4. \tag{14}$$

The equality follows from $\langle f_1|\rho|f_1\rangle = \frac{1}{4}\sum_{a,b}\rho_{ab}$; the bounds follow because $\langle f_1|\rho|f_1\rangle$ is a probability. For the single physical input $f_1 \otimes \omega_E$, set

$$p_f(t) = \langle f_1|\rho_f(t)|f_1\rangle = \Sigma(\rho_f(t))/4, \quad T_F(\rho_f) = 1 - p_f, \quad R(\rho_f) = -\ln p_f. \tag{15}$$

These are the fidelity texture and rugosity conventions used here [1, 2]. At $p_f = 0$, $R = +\infty$; initially $p_f = 1$ and both textures vanish. The texturizing powers evaluated on the unique textureless input are $\Omega_{T_F}(t) = T_F[\Lambda_t(f_1)]$ and $\Omega_R(t) = R[\Lambda_t(f_1)]$.

Since $|+\rangle\langle+| = (I + \sigma^x)/2$, we have $4f_1 = (I + \sigma^x_A)(I + \sigma^x_B)$, and

$$\Sigma(\rho_f) = \mathrm{Tr}[4f_1\rho_f] = 1 + \langle\sigma^x_A\rangle + \langle\sigma^x_B\rangle + \langle\sigma^x_A\sigma^x_B\rangle. \tag{16}$$

A joint x-basis measurement gives $p_f = P^{(x)}_{++}$ directly. Equivalently, apply a Hadamard to each probe and record outcome 00. This uses one measurement setting, whereas full channel reconstruction in the Ising case uses the additional coherence measurements described in Section 3.6.

For any channel, the dual map Λ_t^*, defined by $\mathrm{Tr}[Y\Lambda_t(X)] = \mathrm{Tr}[\Lambda_t^*(Y)X]$, gives the grand-sum response operator

$$\mathcal{R}_t = 4\Lambda_t^*(f_1), \qquad \Sigma[\Lambda_t(X)] = \mathrm{Tr}[4f_1\Lambda_t(X)] = \mathrm{Tr}(\mathcal{R}_t X). \tag{17}$$

These definitions apply in both coupling regimes. Their consequences for each interaction are derived in Sections 3.10 and 5.7.

Part I

Ising coupling

3 Ising coupling: exact generalized dephasing

3.1 Four conditional Hamiltonians

The probe operators S_A^z, S_B^z commute with H_E (which acts only on E) and with V_{ZZ} (which contains only z operators on the probes). Hence $[H^{(\mathrm{ZZ})}, S_A^z] = [H^{(\mathrm{ZZ})}, S_B^z] = 0$. Label each probe basis state by $a = (s_A^a, s_B^a)$ with $s_A^a, s_B^a = \pm 1/2$, and set $P_a = |a\rangle\langle a|$. The projectors satisfy

$$P_a P_b = \delta_{ab} P_a, \qquad \sum_a P_a = I_{AB}, \qquad S_A^z = \sum_a s_A^a P_a, \qquad S_B^z = \sum_a s_B^a P_a. \tag{18}$$

Inserting these resolutions into $H^{(\mathrm{ZZ})} = I_{AB} \otimes H_E + g_A S_A^z \otimes S_1^z + g_B S_B^z \otimes S_N^z$ gives

$$H^{(\mathrm{ZZ})} = \sum_a P_a \otimes H_a, \qquad H_a = H_E + g_A s_A^a S_1^z + g_B s_B^a S_N^z. \tag{19}$$

By induction, $(H^{(\mathrm{ZZ})})^n = \sum_a P_a \otimes H_a^n$: if it holds for n, then $(H^{(\mathrm{ZZ})})^{n+1} = \sum_{a,b} P_a P_b \otimes H_a^n H_b = \sum_a P_a \otimes H_a^{n+1}$. Summing the exponential series term by term,

$$U_t = \sum_a P_a \otimes U_a(t), \qquad U_a(t) = e^{-iH_a t}. \tag{20}$$

This is exact for any bulk Hamiltonian acting only on E, including one with transverse fields.

For equal couplings, the four perturbations are

a	$H_a - H_E$
00	$\frac{g}{2}(S_1^z + S_N^z)$
01	$\frac{g}{2}(S_1^z - S_N^z)$
10	$-\frac{g}{2}(S_1^z - S_N^z)$
11	$-\frac{g}{2}(S_1^z + S_N^z)$

. (21)

These configurations label invariant branches. The input may still coherently occupy all four branches. In general, the differences

$$H_a - H_b = g_A(s_A^a - s_A^b) S_1^z + g_B(s_B^a - s_B^b) S_N^z \tag{22}$$

involve only the two bulk boundary operators.

3.2 Reduced state for an arbitrary probe input

Take an arbitrary probe state and the fixed bulk state,

$$\rho_{AB}(0) = \sum_{a,b} r_{ab} E_{ab}, \qquad \rho_{ABE}(0) = \sum_{a,b} r_{ab}\, E_{ab} \otimes \omega_E. \tag{23}$$

Inserting Eq. (20) on both sides,

$$\rho_{ABE}(t) = U_t \rho_{ABE}(0) U_t^\dagger = \sum_{c,d} \sum_{a,b} r_{ab}\, (P_c E_{ab} P_d) \otimes (U_c \omega_E U_d^\dagger). \tag{24}$$

Since $P_c E_{ab} P_d = \delta_{ca}\delta_{db} E_{ab}$,

$$\rho_{ABE}(t) = \sum_{a,b} r_{ab}\, E_{ab} \otimes U_a(t)\, \omega_E\, U_b(t)^\dagger. \tag{25}$$

The bulk operator multiplying E_{ab} depends on the ordered pair (a, b), although the initial bulk state is the same for all pairs. Taking the partial trace term by term,

$$\rho_{AB}(t) = \sum_{a,b} r_{ab}\, \Gamma_{ab}(t)\, E_{ab}, \qquad \Gamma_{ab}(t) = \mathrm{Tr}_E[U_a(t)\omega_E U_b(t)^\dagger] = \mathrm{Tr}_E[\omega_E U_b(t)^\dagger U_a(t)], \tag{26}$$

where the last form uses cyclicity of the trace on E. In the computational order $(00, 01, 10, 11)$ the full output is

$$\rho_{AB}(t) = \begin{pmatrix} r_{00,00} & r_{00,01}\Gamma_{00,01} & r_{00,10}\Gamma_{00,10} & r_{00,11}\Gamma_{00,11} \\ r_{01,00}\Gamma_{01,00} & r_{01,01} & r_{01,10}\Gamma_{01,10} & r_{01,11}\Gamma_{01,11} \\ r_{10,00}\Gamma_{10,00} & r_{10,01}\Gamma_{10,01} & r_{10,10} & r_{10,11}\Gamma_{10,11} \\ r_{11,00}\Gamma_{11,00} & r_{11,01}\Gamma_{11,01} & r_{11,10}\Gamma_{11,10} & r_{11,11} \end{pmatrix}, \tag{27}$$

where $\Gamma_{aa} = 1$ and $\Gamma_{ba} = \Gamma_{ab}^*$ have been used (Section 3.3). All populations r_{aa} are unchanged; only the six independent complex coherences evolve.

For a pure bulk $\omega_E = |\phi_0\rangle\langle\phi_0|$ and a pure probe input $\sum_a c_a|a\rangle$, the global state is $\sum_a c_a|a\rangle \otimes U_a|\phi_0\rangle$ and $\Gamma_{ab} = \langle\phi_0|U_b^\dagger U_a|\phi_0\rangle$ is the complex overlap of two conditional bulk histories. Its squared modulus is a conditional Loschmidt echo; the complex amplitude, including its phase, is needed for the channel.

3.3 Schur form and its physical constraints

Since Eq. (26) holds for every input, the channel acts on each matrix unit as

$$\boxed{\Lambda_t(E_{ab}) = \Gamma_{ab}(t)E_{ab}, \qquad \Lambda_t(X) = \Gamma(t) \circ X,} \tag{28}$$

where $\circ$ is the entrywise Schur (Hadamard) product and $\Gamma(t) = (\Gamma_{ab}(t))_{a,b=1}^4$. The sixteen E_{ab} are superoperator eigenoperators, with eigenvalues Γ_{ab}, four of which equal one.

The matrix Γ is a correlation matrix:

$$\Gamma = \Gamma^\dagger \succeq 0, \qquad \Gamma_{aa} = 1, \qquad |\Gamma_{ab}| \le 1, \qquad \Gamma_{ab}(0) = 1. \tag{29}$$

Proof. Set $A_a(t) = U_a(t)\sqrt{\omega_E}$, so that $\Gamma_{ab} = \mathrm{Tr}_E(A_a A_b^\dagger)$. For any $v \in \mathbb{C}^4$,

$$v^\dagger \Gamma v = \sum_{a,b} v_a^* \Gamma_{ab} v_b = \mathrm{Tr}_E\Big[\Big(\sum_a v_a^* A_a\Big)\Big(\sum_b v_b^* A_b\Big)^\dagger\Big] \ge 0, \tag{30}$$

because $(\sum_b v_b^* A_b)^\dagger = \sum_b v_b A_b^\dagger$ and $\mathrm{Tr}(YY^\dagger) \ge 0$. Hence $\Gamma = \Gamma^\dagger \succeq 0$. Unitarity and $\mathrm{Tr}\,\omega_E = 1$ give $\Gamma_{aa} = \mathrm{Tr}_E(U_a\omega_E U_a^\dagger) = 1$. Nonnegativity of each 2×2 principal minor gives $\Gamma_{aa}\Gamma_{bb} - |\Gamma_{ab}|^2 = 1 - |\Gamma_{ab}|^2 \ge 0$. Finally $U_a(0) = I$ gives $\Gamma_{ab}(0) = 1$. Equivalently, for a decomposition $\omega_E = \sum_\nu p_\nu|\phi_\nu\rangle\langle\phi_\nu|$ and an output basis $\{|r\rangle\}$, the numbers $k_{r\nu,a} = \sqrt{p_\nu}\langle r|U_a|\phi_\nu\rangle$ give the Gram factorization $\Gamma_{ab} = \sum_{r,\nu} k_{r\nu,a} k_{r\nu,b}^*$. □

Conversely, every positive matrix with unit diagonal defines a CPTP Schur map, by the Kraus construction of Eq. (42). This converse characterizes the map class, not which members a specified XXZ Hamiltonian realizes at a given time.

All four populations are fixed and $\Lambda_t(I_4) = \sum_a \Gamma_{aa} E_{aa} = I_4$, so the channel is unital. Six complex off-diagonal factors specify the nontrivial action, subject to positivity. Because the superoperator is diagonal in the Hilbert–Schmidt orthonormal basis $\{E_{ab}\}$, its singular values are $|\Gamma_{ab}|$. The common interacting bulk can correlate the two probes, so a product of independent single-probe dephasing channels cannot be assumed. There is no general monotonicity of $|\Gamma_{ab}(t)|$.

3.4 Exact expressions for the coherence factors and their derivatives

For any convex decomposition $\omega_E = \sum_\nu p_\nu |\phi_\nu\rangle\langle\phi_\nu|$, Eq. (26) reads

$$\Gamma_{ab}(t) = \sum_\nu p_\nu \langle\phi_\nu|U_b(t)^\dagger U_a(t)|\phi_\nu\rangle. \tag{31}$$

With $\dot{U}_a = -iH_aU_a$ and $\dot{U}_b^\dagger = iU_b^\dagger H_b$ (both using $[H_a, U_a] = [H_b, U_b] = 0$),

$$\dot{\Gamma}_{ab} = \mathrm{Tr}_E[\omega_E(iU_b^\dagger H_b)U_a + \omega_E U_b^\dagger(-iH_aU_a)] = -i\,\mathrm{Tr}_E[\omega_E U_b(t)^\dagger(H_a - H_b)U_a(t)]. \tag{32}$$

With Eq. (22),

$$\dot{\Gamma}_{ab} = -i\Big[g_A(s_A^a - s_A^b)\,\mathrm{Tr}_E(\omega_E U_b^\dagger S_1^z U_a) + g_B(s_B^a - s_B^b)\,\mathrm{Tr}_E(\omega_E U_b^\dagger S_N^z U_a)\Big]. \tag{33}$$

Only the two boundary operators enter. No numerical time differentiation is required once the conditional states and the matrix elements of S_1^z and S_N^z are available.

At $t = 0$, $U_a(0) = I$ gives

$$\Gamma_{ab}(0) = 1, \qquad \dot{\Gamma}_{ab}(0) = -i(\varepsilon_a - \varepsilon_b), \qquad \varepsilon_a = \mathrm{Tr}_E(\omega_E H_a) \in \mathbb{R}. \tag{34}$$

The initial derivative is purely imaginary. Its consequence for the canonical generator is derived in Section 4.9.

3.5 Dephasing and depolarization

A depolarizing map has the form $\mathcal{D}_\lambda(X) = \lambda X + (1-\lambda)\,\mathrm{Tr}(X)I_4/4$. The Ising map fixes every P_a. Requiring $\mathcal{D}_\lambda(P_a) = \lambda P_a + (1-\lambda)I_4/4 = P_a$ gives $(1-\lambda)(I_4/4 - P_a) = 0$, which forces $\lambda = 1$. Thus a nonidentity Ising Schur map cannot be a depolarizing map. Dephasing includes coherent diagonal rotations and revivals; it does not imply irreversible or strictly dissipative evolution.

3.6 One output determines the complete channel

For the texture preparation f_1 of Eq. (13), with full output state $\rho_f(t) = \Lambda_t(f_1)$, the Schur form gives $[\rho_f(t)]_{ab} = \Gamma_{ab}(t)(f_1)_{ab} = \Gamma_{ab}(t)/4$, hence

$$\boxed{\rho_f(t) = \Gamma(t)/4, \qquad \Gamma(t) = 4\rho_f(t), \qquad \Lambda_t(X) = [4\rho_f(t)] \circ X.} \tag{35}$$

Consequently the *entire density matrix* of one known output determines every property of the instantaneous Ising channel. Temporal properties require these outputs at the relevant times. A single scalar texture value cannot replace this matrix. This identity is special to the Ising Schur family; it does not hold for arbitrary two-probe channels.

The populations of ρ_f are $1/4$, so its 12 remaining real parameters are its six complex coherences. In a Pauli expansion, $\langle ZI\rangle = \langle IZ\rangle = \langle ZZ\rangle = 0$. The eight local measurement settings $(\alpha, \beta) \neq (z, z)$, $\alpha, \beta \in \{x, y, z\}$, recover the other twelve Pauli coefficients, including single-spin expectations from their marginals. Thus one input preparation together with output tomography suffices.

The time-local data of Section 4 follow from the same output and its time derivative:

$$\dot{\Gamma}(t) = 4\dot{\rho}_f(t), \qquad \eta_{ab}(t) = \frac{\dot{\Gamma}_{ab}(t)}{\Gamma_{ab}(t)} = \frac{[\dot{\rho}_f(t)]_{ab}}{[\rho_f(t)]_{ab}} \quad (a \neq b,\ [\rho_f(t)]_{ab} \neq 0). \tag{36}$$

3.7 Choi, Kraus and spectral consequences

Inserting Eq. (28) into Eq. (12),

$$J_t = \frac{1}{4}\sum_{a,b}\Gamma_{ab}\,E_{ab}\otimes E_{ab} = \frac{1}{4}\sum_{a,b}\Gamma_{ab}|aa\rangle\langle bb|. \tag{37}$$

Let $W:\mathcal{H}_{AB}\to\mathcal{H}_{AB}\otimes\mathcal{H}_R$ be the isometry $W|a\rangle = |aa\rangle$. Then $W(\Gamma/4)W^\dagger = \frac{1}{4}\sum_{a,b}\Gamma_{ab}|aa\rangle\langle bb|$, so

$$J_t = W\frac{\Gamma(t)}{4}W^\dagger = W\rho_f(t)W^\dagger. \tag{38}$$

Since $W^\dagger W = I_4$, conjugation by W preserves nonzero spectra. The Choi support has dimension at most four. In particular,

$$\operatorname{rank} J_t = \operatorname{rank}\Gamma \le 4, \tag{39}$$

$$\operatorname{spec}(J_t) = \operatorname{spec}(\Gamma/4) \text{ with twelve additional zeros}, \tag{40}$$

$$S(J_t) = S(\rho_f), \qquad \operatorname{Tr} J_t^2 = \operatorname{Tr}\rho_f^2 = \frac{1}{16}\sum_{a,b}|\Gamma_{ab}|^2. \tag{41}$$

If $\Gamma = \sum_k \lambda_k v_k v_k^\dagger$ with $\lambda_k \ge 0$ and normalized eigenvectors v_k, one minimal Kraus family is

$$D_k = \sqrt{\lambda_k}\,\operatorname{diag}(v_{k,1},\dots,v_{k,4}), \qquad \Lambda_t(X) = \sum_k D_k X D_k^\dagger. \tag{42}$$

Indeed $\sum_k D_k E_{ab} D_k^\dagger = \sum_k \lambda_k v_{k,a} v_{k,b}^* E_{ab} = \Gamma_{ab}E_{ab}$, and $\sum_k D_k^\dagger D_k = \operatorname{diag}(\Gamma_{aa}) = I_4$. Choi rank and superoperator rank are distinct: the identity has Choi rank one and superoperator rank sixteen.

The Kraus decomposition diagonalizes the positive 4×4 matrix Γ. The canonical time-local generator of Section 4 requires a *different* diagonalization, that of a 3×3 decoherence matrix built from logarithmic derivatives of Γ. The eigenvalues of Γ are not the canonical decoherence rates.

Unitarity. The map is unitary iff $\operatorname{rank}\Gamma = 1$. If $\operatorname{rank}\Gamma = 1$, then $\Gamma = \lambda vv^\dagger$; unit diagonal gives $\lambda|v_a|^2 = 1$, so $u_a = \sqrt{\lambda}\,v_a$ obeys $|u_a| = 1$ and $\Gamma_{ab} = u_a u_b^*$. Then $\Lambda_t(X) = \operatorname{diag}(u)X\operatorname{diag}(u)^\dagger$. Conversely, a unitary channel has a pure Choi state, and by Eq. (38) $\operatorname{rank}\Gamma = \operatorname{rank} J_t = 1$. This does not require that the unitary factor into two local rotations.

Reference entanglement. Take the partial transpose of J_t on R: $|aa\rangle\langle bb| \mapsto |ab\rangle\langle ba|$, so $J_t^{T_R} = \frac{1}{4}\sum_{a,b}\Gamma_{ab}|ab\rangle\langle ba|$. The terms with $a = b$ give the eigenvalue $1/4$ four times, on $|aa\rangle$. For each pair $a < b$, the two-dimensional block on $\{|ab\rangle, |ba\rangle\}$ is

$$\frac{1}{4}\begin{pmatrix} 0 & \Gamma_{ab} \\ \Gamma_{ab}^* & 0 \end{pmatrix}, \qquad \text{eigenvalues } \pm\frac{|\Gamma_{ab}|}{4}. \tag{43}$$

Hence its reference–output negativity is

$$\mathcal{N}_{AB|R}(J_t) = \frac{1}{4}\sum_{a<b}|\Gamma_{ab}|. \tag{44}$$

This is distinct from physical $A|B$ entanglement. The Ising map is entanglement breaking iff all off-diagonal Γ_{ab} vanish: otherwise its Choi state has a negative partial transpose and is entangled, while the diagonal case has the explicitly separable Choi state $\frac{1}{4}\sum_a |aa\rangle\langle aa|$.

3.8 Invertibility, time-local generator and intermediate maps

Because the superoperator is diagonal on $\{E_{ab}\}$,

$$\det \Lambda_t = \prod_{a,b} \Gamma_{ab}(t) = \prod_{a<b} \Gamma_{ab}\Gamma_{ba} = \prod_{a<b} |\Gamma_{ab}(t)|^2, \tag{45}$$

using $\Gamma_{aa} = 1$ and $\Gamma_{ba} = \Gamma_{ab}^*$. Therefore

$$\Lambda_t \text{ invertible} \iff \Gamma_{ab}(t) \neq 0 \quad \forall a < b, \qquad \Lambda_t^{-1}(E_{ab}) = \Gamma_{ab}(t)^{-1} E_{ab}. \tag{46}$$

At such times the exact time-local generator $\mathcal{L}_t = \dot{\Lambda}_t \Lambda_t^{-1}$ acts as

$$\mathcal{L}_t(E_{ab}) = \frac{\dot{\Gamma}_{ab}(t)}{\Gamma_{ab}(t)} E_{ab} \equiv \eta_{ab}(t) E_{ab}. \tag{47}$$

If Λ_s is invertible, the unique intermediate linear map $\Lambda_{t,s}$ with $\Lambda_t = \Lambda_{t,s}\Lambda_s$ is $\Lambda_{t,s} = \Lambda_t \Lambda_s^{-1}$, i.e. Schur multiplication by $Q_{ab}(t,s) = \Gamma_{ab}(t)/\Gamma_{ab}(s)$. The matrix Q is Hermitian, since $Q_{ba} = \Gamma_{ab}^*(t)/\Gamma_{ab}^*(s) = Q_{ab}^*$, and has unit diagonal. A Schur map with unit diagonal is trace preserving, and it is completely positive iff its multiplier is positive semidefinite (its normalized Choi state is $WQW^\dagger/4$). Hence

$$\Lambda_{t,s} \text{ is CPTP} \quad \iff \quad Q(t,s) \succeq 0. \tag{48}$$

At singular times the division is undefined and this test cannot be applied.

Lemma 3.1. *On an interval where the coherence factors obey $\dot{\Gamma}_{ab} = \eta_{ab}\Gamma_{ab}$ with locally integrable η_{ab}, no Γ_{ab} can vanish at a finite time if it is nonzero at the start of the interval.*

Proof. The linear equation integrates to $\Gamma_{ab}(t) = \Gamma_{ab}(s) \exp \int_s^t \eta_{ab}(\tau)\, d\tau$, and the exponential of a finite integral is nonzero. Equivalently, $\det \Lambda_t = \det \Lambda_s \exp \int_s^t \operatorname{Tr} \mathcal{L}_\tau \, d\tau$. □

Thus a finite, locally integrable time-local generator cannot take an invertible evolution to a singular channel in finite time. If a coherence factor vanishes at s and later becomes nonzero, no linear map $\Lambda_{t,s}$ on the probes can satisfy $\Lambda_t = \Lambda_{t,s}\Lambda_s$, because $\Lambda_s(E_{ab}) = 0$ while $\Lambda_t(E_{ab}) \neq 0$. This precludes divisibility through that time. The global unitary and the reduced CPTP maps Λ_t themselves remain analytic in t.

3.9 Probe entanglement from the same input

For two-qubit states, concurrence C and negativity $\mathcal{N}$ test entanglement of A with B; their definitions are in Appendix A. For a Schur channel, the output from $|{+}{+}\rangle$ also determines the maximum concurrence generated from separable probe states.

Lemma 3.2 (Local-filter identity)**.** *For 2×2 matrices A, B and a positive two-qubit operator ρ, with concurrence defined from the eigenvalues of $\rho\widetilde{\rho}$ without normalization, $C[(A\otimes B)\rho(A\otimes B)^\dagger] = |\det A \det B|\, C(\rho)$.*

Proof. For any 2×2 matrix, $A^{\mathsf{T}} \sigma_y A = \det(A)\, \sigma_y$. Taking the complex conjugate and using $\sigma_y^* = -\sigma_y$ gives $A^\dagger \sigma_y A^* = \det(A)^* \sigma_y$. For invertible A, the first identity also gives $\sigma_y A^{\mathsf{T}} \sigma_y = \det(A) A^{-1}$. Let $\rho' = (A \otimes B)\rho(A \otimes B)^\dagger$ and $Y = \sigma_y \otimes \sigma_y$. Then

$$\begin{aligned} \rho'\widetilde{\rho'} &= (A \otimes B)\rho(A \otimes B)^\dagger Y (A \otimes B)^* \rho^* (A \otimes B)^{\mathsf{T}} Y \\ &= \det(A)^* \det(B)^* \, (A \otimes B)\rho\, Y \rho^* Y \, Y (A \otimes B)^{\mathsf{T}} Y \\ &= |\det A \det B|^2 \, (A \otimes B)\, \rho\widetilde{\rho}\, (A \otimes B)^{-1}. \end{aligned} \tag{49}$$

This is similar to $|\det A \det B|^2 \rho\widetilde{\rho}$, so the square roots of its eigenvalues, and hence C, scale by $|\det A \det B|$. Singular filters follow by continuity of C. □

Proposition 3.3. *For a fixed Ising Schur channel, its maximum output concurrence over separable probe inputs is*

$$\max_{\rho\in\mathrm{Sep}(A:B)} C[\Lambda_t(\rho)] = C[\rho_f(t)] = C[\Gamma(t)/4]. \tag{50}$$

Proof. For a normalized pure product input $|\psi\rangle = (a_0|0\rangle + a_1|1\rangle)\otimes(b_0|0\rangle + b_1|1\rangle)$, the components are $\psi_{ij} = a_i b_j$. By Eq. (28), $\Lambda_t(|\psi\rangle\langle\psi|)_{ab} = \psi_a \Gamma_{ab} \psi_b^*$, i.e. $\Lambda_t(|\psi\rangle\langle\psi|) = D_\psi \Gamma D_\psi^\dagger$ with $D_\psi = \mathrm{diag}(\psi)$. Using $\Gamma = 4\rho_f$ and $2D_\psi = A\otimes B$ with $A = \sqrt{2}\,\mathrm{diag}(a_0, a_1)$, $B = \sqrt{2}\,\mathrm{diag}(b_0, b_1)$,

$$\Lambda_t(|\psi\rangle\langle\psi|) = (A\otimes B)\rho_f(A\otimes B)^\dagger. \tag{51}$$

This output has unit trace because ρ_f has uniform diagonal $1/4$: its trace is $\sum_{ij} 4|a_i|^2|b_j|^2/4 = 1$. Lemma 3.2 gives $C_{\mathrm{out}} = |\det A \det B| C(\rho_f) = 4|a_0 a_1 b_0 b_1| C(\rho_f) \le C(\rho_f)$, since $|a_0 a_1| \le \frac{1}{2}(|a_0|^2 + |a_1|^2) = \frac{1}{2}$ and likewise for b. A separable input is a convex combination $\rho = \sum_i q_i |\psi_i\rangle\langle\psi_i|$ of pure product states. Linearity of Λ_t and convexity of C give $C[\Lambda_t(\rho)] \le \sum_i q_i C[\Lambda_t(|\psi_i\rangle\langle\psi_i|)] \le C(\rho_f)$. The input $|{+}{+}\rangle$ attains equality. □

In particular, the Ising map can create probe entanglement from some separable input iff $\rho_f(t)$ is entangled. Texture and Choi reference–output negativity are different quantities and do not replace this criterion.

3.10 Ising texture and channel fidelities

Using $\rho_f = \Gamma/4$ in Eq. (15), together with $\sum_{a,b}\Gamma_{ab} = 4 + \sum_{a<b}(\Gamma_{ab} + \Gamma_{ab}^*) = 4 + 2\sum_{a<b}\Re\Gamma_{ab}$,

$$p_f = \frac{1}{16}\sum_{a,b}\Gamma_{ab} = \frac{1}{4} + \frac{1}{8}\sum_{a<b}\Re\Gamma_{ab}, \tag{52}$$

$$\Omega_{T_F} = \frac{3}{4} - \frac{1}{8}\sum_{a<b}\Re\Gamma_{ab}, \qquad \Omega_R = -\ln\left(\frac{1}{16}\sum_{a,b}\Gamma_{ab}\right). \tag{53}$$

For the same Schur channel, the entanglement fidelity relative to the identity is

$$F_e = \langle\Phi_4|J_t|\Phi_4\rangle = \frac{1}{4}\sum_{c,d}\langle cc|\Big(\frac{1}{4}\sum_{a,b}\Gamma_{ab}|aa\rangle\langle bb|\Big)|dd\rangle = \frac{1}{16}\sum_{a,b}\Gamma_{ab} = p_f. \tag{54}$$

The general relation $F_{\mathrm{avg}} = (dF_e + 1)/(d+1)$ between average and entanglement fidelities [3], with $d = 4$, then gives

$$\boxed{F_e = p_f = \Sigma(\rho_f)/4, \qquad F_{\mathrm{avg}} = \frac{4F_e + 1}{5} = \frac{1 + \Sigma(\rho_f)}{5}.} \tag{55}$$

The equality $F_e = p_f$ uses the Schur structure. These are fidelities of the *two-probe map relative to the identity*, not fidelities of sending an unknown state from A to B.

Only the real sum of the six coherences enters p_f. It does not fix their individual magnitudes or phases and does not determine purity, probe entanglement, or CP divisibility. Coherent phase evolution can change texture even when the map is unitary.

3.11 Response operator and texture preservation

In the Ising case, Eq. (17) reads $\Sigma[\Lambda_t(X)] = \sum_{a,b}\Gamma_{ab}X_{ab}$, while $\mathrm{Tr}(\mathcal{R}_t X) = \sum_{a,b}(\mathcal{R}_t)_{ba}X_{ab}$. Comparing coefficients of X_{ab} gives $(\mathcal{R}_t)_{ba} = \Gamma_{ab}$, i.e.

$$\mathcal{R}_t = \Gamma(t)^{\mathsf{T}}. \tag{56}$$

The transpose is ordinary, not a Hermitian conjugate; since Γ is Hermitian, $\Gamma^{\mathsf{T}} = \Gamma^*$. In general, fixing the free state, $\Lambda_t(f_1) = f_1$, and preserving the grand sum for every input, $\Lambda_t^*(f_1) = f_1$, are different conditions [2]. For this Schur family, $\Lambda_t(f_1) = \Gamma/4$ and $\Lambda_t^*(f_1) = \Gamma^{\mathsf{T}}/4$, so each condition forces $\Gamma_{ab} = 1$ for all a, b, hence the identity channel. Moreover $\Omega_{T_F} = 0$ means $p_f = 1$, i.e. $\sum_{a<b} \Re\Gamma_{ab} = 6$; with $|\Gamma_{ab}| \leq 1$ this forces every $\Gamma_{ab} = 1$. Consequently $\Omega_{T_F} = 0$ also certifies the identity within the known Ising family.

3.12 Summary of the Ising channel structure

Six complex overlaps between conditional bulk evolutions determine the Ising channel. The output $\rho_f = \Gamma/4$ therefore specifies its full action, including its Choi state, fidelities and ability to generate probe entanglement. Evaluating the overlaps requires the conditional dynamics for the chosen J, Δ, g_A, g_B, N and ω_E.

4 Ising coupling: canonical form of the time-dependent dephasing channel

The Schur structure of Eq. (28) fixes the time-local generator in closed form. Here we bring it to the canonical form of Hall *et al.* [4], which uniquely fixes the canonical rates. The associated operators retain phase and degeneracy freedoms, as specified in Section 4.8. Every result of this section holds for an arbitrary fixed bulk preparation ω_E and applies to every probe input $\rho_{AB}(0)$, pure or mixed, product or entangled: the canonical form is a property of the channel Λ_t, which enters only through the coherence factors $\Gamma_{ab}(t)$ of Eq. (26).

The notation distinguishes three quantities: $\Gamma_{ab}(t)$ is a complex channel multiplier, $\kappa_{ab}(t)$ is its real logarithmic decay rate, and $\gamma_k(t)$ is a real canonical rate. The pair of configuration indices a, b labels a coherence; the single index k labels an eigenmode of the decoherence matrix. Both rates have units of inverse time, whereas Γ and Γ_{ab} are dimensionless.

4.1 Time-local generator of the Schur channel

When the dynamical map is invertible, the reduced evolution obeys a time-local master equation with generator $\mathcal{L}_t = \dot{\Lambda}_t \Lambda_t^{-1}$ [4]. For the Schur channel this requires $\Gamma_{ab}(t) \neq 0$ for all a, b (Section 3.8). Differentiating $\rho_{ab}(t) = \Gamma_{ab}(t) r_{ab}$ and, for $\Gamma_{ab}(t) \neq 0$, substituting $r_{ab} = \rho_{ab}(t)/\Gamma_{ab}(t)$ gives

$$\dot{\rho}_{ab}(t) = \eta_{ab}(t)\, \rho_{ab}(t), \qquad \eta_{ab}(t) = \frac{\dot{\Gamma}_{ab}(t)}{\Gamma_{ab}(t)} = \frac{d}{dt} \ln \Gamma_{ab}(t), \tag{57}$$

so $\mathcal{L}_t(X)_{ab} = \eta_{ab}(t) X_{ab}$, in agreement with Eq. (47). The constraints of Eq. (29) imply $\eta_{aa} = 0$ and $\eta_{ba} = \eta_{ab}^*$, so the 4×4 matrix $\eta = (\eta_{ab})$ is Hermitian with zero diagonal. All generator data are six complex logarithmic derivatives. Writing each nonzero factor in polar form, $\Gamma_{ab} = R_{ab} e^{i\theta_{ab}}$ with $R_{ab} > 0$,

$$\dot{\Gamma}_{ab} = (\dot{R}_{ab} + i R_{ab} \dot{\theta}_{ab}) e^{i\theta_{ab}}, \qquad \eta_{ab} = \frac{\dot{R}_{ab}}{R_{ab}} + i\dot{\theta}_{ab}. \tag{58}$$

We therefore define, without assuming Markovianity,

$$\eta_{ab} = -\kappa_{ab} + i\omega_{ab}, \qquad \kappa_{ab} = -\frac{d}{dt} \ln|\Gamma_{ab}|, \qquad \omega_{ab} = \frac{d}{dt} \arg \Gamma_{ab}. \tag{59}$$

Here $\kappa_{ab} = \kappa_{ba}$ and $\omega_{ab} = -\omega_{ba}$; the derivative ω_{ab} is taken on a locally continuous phase branch. Equivalently, $\omega_{ab} = \Im(\dot{\Gamma}_{ab}/\Gamma_{ab})$, which avoids artificial jumps of the principal argument. For any

input with $r_{ab} \neq 0$,

$$\kappa_{ab}(t) = -\frac{1}{|\rho_{ab}(t)|}\frac{d|\rho_{ab}(t)|}{dt}, \tag{60}$$

so κ_{ab} is the fractional rate of loss of the coherence between configurations a and b, independent of the input amplitude. It is an observed coherence damping rate, not in general an eigenvalue of the canonical decoherence matrix. Over an interval on which Γ_{ab} does not vanish, $\int_{t_1}^{t_2} \kappa_{ab}\, dt = \ln(|\Gamma_{ab}(t_1)|/|\Gamma_{ab}(t_2)|)$.

4.2 Canonical form

Let $\{O_j\}_{j=1}^{15}$ be an orthonormal basis of traceless probe operators, $\mathrm{Tr}\, O_j = 0$ and $\mathrm{Tr}(O_j^\dagger O_k) = \delta_{jk}$. Any time-local master equation of a d-dimensional system can be written as [4]

$$\dot\rho = -i[H(t), \rho] + \sum_{j,k} \mathsf{d}_{jk}(t)\Big(O_j \rho O_k^\dagger - \tfrac{1}{2}\{O_k^\dagger O_j, \rho\}\Big), \tag{61}$$

with a Hermitian traceless $H(t)$ and a Hermitian decoherence matrix $\mathsf{d}(t)$. Diagonalizing d gives

$$\dot\rho = -i[H(t), \rho] + \sum_{k} \gamma_k(t)\Big(L_k(t) \rho L_k^\dagger(t) - \tfrac{1}{2}\{L_k^\dagger(t) L_k(t), \rho\}\Big), \tag{62}$$

where the $L_k(t)$ form an orthonormal basis of traceless operators. The canonical decoherence rates $\gamma_k(t)$ are the eigenvalues of $\mathsf{d}(t)$; they are uniquely determined, invariant under the choice of the basis $\{O_j\}$, time dependent, and can be negative [4]. The Hamiltonian $H(t)$ acts on the probe pair and is distinct from the total probe–bulk Hamiltonian $H^{(\mathrm{ZZ})}$. We construct this form explicitly for $\mathcal{L}_t$ of Eq. (57).

4.3 Diagonal operator basis

Define the three diagonal, traceless and Hermitian operators

$$F_1 = \frac{\sigma_A^z \otimes I}{2}, \qquad F_2 = \frac{I \otimes \sigma_B^z}{2}, \qquad F_3 = \frac{\sigma_A^z \otimes \sigma_B^z}{2}. \tag{63}$$

Each is a product of Paulis divided by two, so $\mathrm{Tr}\, F_\mu = 0$ and $\mathrm{Tr}(F_\mu F_\nu) = \delta_{\mu\nu}$. For the configuration $a = (s_A^a, s_B^a)$, $F_\mu|a\rangle = g_\mu^a|a\rangle$ with $\mathbf{g}^a = (g_1^a, g_2^a, g_3^a) = (s_A^a,\, s_B^a,\, 2s_A^a s_B^a)$. Collecting these eigenvalues into the 4×3 matrix $G = (g_\mu^a)_{a\mu}$,

$$G = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}, \qquad \boxed{G^{\mathsf{T}} G = I_3, \qquad G^{\mathsf{T}}\mathbf{1} = 0, \qquad G G^{\mathsf{T}} = I_4 - \tfrac{1}{4}\mathbf{1}\mathbf{1}^{\mathsf{T}},} \tag{64}$$

where $\mathbf{1} = (1,1,1,1)^{\mathsf{T}}$. The first two identities follow by direct multiplication: each column has four entries $\pm\frac{1}{2}$, distinct columns differ in sign at exactly two rows, and each column has two entries of each sign. The third follows because the three columns and $\mathbf{1}/2$ form an orthonormal basis of $\mathbb{R}^4$. Row a of G, viewed as a column vector, is $\mathbf{g}^a$; hence $\sum_a \mathbf{g}^a = G^{\mathsf{T}}\mathbf{1} = 0$.

Complete $\{F_1, F_2, F_3\}$ with any twelve orthonormal traceless off-diagonal operators to obtain a basis $\{O_j\}$. We take the ansatz of a diagonal Hamiltonian

$$H_{\mathrm{eff}} = \sum_a h_a P_a, \qquad h_a \in \mathbb{R}, \qquad \sum_a h_a = 0, \tag{65}$$

where the last condition fixes the irrelevant scalar gauge, and a decoherence matrix supported on the diagonal operators,

$$\mathsf{d} = S + iA, \qquad S = S^{\mathsf{T}} \in \mathbb{R}^{3\times 3}, \qquad A = -A^{\mathsf{T}} \in \mathbb{R}^{3\times 3}, \tag{66}$$

indexed by $\mu, \nu \in \{1, 2, 3\}$, with all other entries zero. Any Hermitian 3×3 matrix has this decomposition, with nine real parameters. *No positivity assumption* is imposed on d or its eigenvalues: these may have either sign while Λ_t remains CPTP. The ansatz reads

$$\mathcal{L}_t[X] = -i[H_{\text{eff}}, X] + \sum_{\mu,\nu=1}^{3} \mathsf{d}_{\mu\nu}\Big(F_\mu X F_\nu - \tfrac{1}{2}\{F_\nu F_\mu, X\}\Big). \tag{67}$$

4.4 Action of each term on a matrix unit

Because every F_μ and H_{eff} is diagonal,

$$F_\mu E_{ab} F_\nu = g^a_\mu g^b_\nu E_{ab}, \qquad F_\nu F_\mu E_{ab} = g^a_\nu g^a_\mu E_{ab}, \qquad E_{ab} F_\nu F_\mu = g^b_\nu g^b_\mu E_{ab}, \tag{68}$$

$$[H_{\text{eff}}, E_{ab}] = (h_a - h_b) E_{ab}. \tag{69}$$

Thus the ansatz maps E_{ab} to a multiple of itself, with the exact scalar coefficient

$$\eta_{ab} = -i(h_a - h_b) + \sum_{\mu\nu} \mathsf{d}_{\mu\nu} g^a_\mu g^b_\nu - \frac{1}{2} \sum_{\mu\nu} \mathsf{d}_{\mu\nu} (g^a_\nu g^a_\mu + g^b_\nu g^b_\mu). \tag{70}$$

In vector notation,

$$\boxed{\eta_{ab} = -i(h_a - h_b) + \mathbf{g}^{a\mathsf{T}} \mathsf{d}\, \mathbf{g}^b - \tfrac{1}{2} \mathbf{g}^{a\mathsf{T}} \mathsf{d}\, \mathbf{g}^a - \tfrac{1}{2} \mathbf{g}^{b\mathsf{T}} \mathsf{d}\, \mathbf{g}^b.} \tag{71}$$

This must equal $\dot{\Gamma}_{ab}/\Gamma_{ab}$ for every a, b. Because the $\mathbf{g}^a$ are real and A is real antisymmetric, $\mathbf{x}^{\mathsf{T}} A \mathbf{x} = 0$ for every real $\mathbf{x}$, so $\mathbf{g}^{a\mathsf{T}} \mathsf{d} \mathbf{g}^a = \mathbf{g}^{a\mathsf{T}} S \mathbf{g}^a$ is real and $\mathbf{g}^{a\mathsf{T}} \mathsf{d} \mathbf{g}^b = \mathbf{g}^{a\mathsf{T}} S \mathbf{g}^b + i\, \mathbf{g}^{a\mathsf{T}} A \mathbf{g}^b$. Separating real and imaginary parts, with $\Delta_{ab} = \mathbf{g}^a - \mathbf{g}^b$,

$$\Re \eta_{ab} = \mathbf{g}^{a\mathsf{T}} S \mathbf{g}^b - \tfrac{1}{2} \mathbf{g}^{a\mathsf{T}} S \mathbf{g}^a - \tfrac{1}{2} \mathbf{g}^{b\mathsf{T}} S \mathbf{g}^b = -\tfrac{1}{2} \Delta^{\mathsf{T}}_{ab} S\, \Delta_{ab}, \tag{72}$$

$$\Im \eta_{ab} = -(h_a - h_b) + \mathbf{g}^{a\mathsf{T}} A\, \mathbf{g}^b. \tag{73}$$

Comparing with Eq. (59),

$$\kappa_{ab} = \tfrac{1}{2} \Delta^{\mathsf{T}}_{ab} S\, \Delta_{ab}, \tag{74}$$

$$\omega_{ab} = -(h_a - h_b) + \mathbf{g}^{a\mathsf{T}} A\, \mathbf{g}^b. \tag{75}$$

The symmetric part governs instantaneous damping; the antisymmetric part contributes phases beyond those of a diagonal Hamiltonian.

4.5 Matrix form and unique inversion

Define $q_a = \frac{1}{2} \mathbf{g}^{a\mathsf{T}} S \mathbf{g}^a$ and the vectors $h = (h_a)_a$, $q = (q_a)_a$. Equation (71) for all sixteen entries, including $a = b$ where it gives $\eta_{aa} = 0$, is the matrix identity

$$\eta = G \mathsf{d} G^{\mathsf{T}} + (-ih - q)\mathbf{1}^{\mathsf{T}} + \mathbf{1}(ih - q)^{\mathsf{T}}, \tag{76}$$

with ordinary transposes: the (a, b) entry of the right side is $\mathbf{g}^{a\mathsf{T}} \mathsf{d} \mathbf{g}^b - ih_a - q_a + ih_b - q_b$. Multiplying on the left by G^{T} and on the right by G, and using Eq. (64),

$$G^{\mathsf{T}} \eta G = (G^{\mathsf{T}} G) \mathsf{d} (G^{\mathsf{T}} G) + G^{\mathsf{T}}(-ih - q)(\mathbf{1}^{\mathsf{T}} G) + (G^{\mathsf{T}} \mathbf{1})(ih - q)^{\mathsf{T}} G = \mathsf{d}, \tag{77}$$

that is,

$$\boxed{\mathsf{d}(t) = G^{\mathsf{T}}\eta(t)G, \qquad \mathsf{d}_{\mu\nu} = \sum_{a,b} g^a_\mu\, \eta_{ab}\, g^b_\nu, \qquad S = \Re\mathsf{d}, \quad A = \Im\mathsf{d}.} \tag{78}$$

This reconstructs all nine real parameters of d. Its antisymmetric part is $A_{\mu\nu} = \sum_{a,b} g^a_\mu \omega_{ab} g^b_\nu$, since g is real.

For the Hamiltonian, sum Eq. (75) over b at fixed a:

$$\sum_b \omega_{ab} = -4h_a + \sum_b h_b + \mathbf{g}^{a\mathsf{T}} A \sum_b \mathbf{g}^b = -4h_a, \tag{79}$$

using $\sum_b h_b = 0$ and $\sum_b \mathbf{g}^b = 0$. Therefore

$$\boxed{h_a(t) = -\frac{1}{4}\sum_b \omega_{ab}(t) = -\frac{1}{4}\sum_b \Im\eta_{ab}(t), \qquad H_{\text{eff}}(t) = \sum_a h_a(t) P_a.} \tag{80}$$

The four diagonal traceless matrices are spanned by F_1, F_2, F_3, so H_{eff} may contain σ^z_A, σ^z_B and an induced $\sigma^z_A\sigma^z_B$ term.

Uniqueness. The map $(\mathsf{d}, h) \mapsto \eta$ of Eq. (76), with d Hermitian 3×3 (nine real parameters) and h real with $\sum_a h_a = 0$ (three real parameters), is real-linear and takes values in the Hermitian 4×4 matrices with zero diagonal (six complex, or twelve real, parameters). It is injective: if $\eta = 0$, then Eq. (78) gives $\mathsf{d} = 0$, and Eq. (80) gives $h = 0$. An injective linear map between real spaces of equal dimension twelve is bijective. Hence for every Hermitian η with zero diagonal there is exactly one pair $(\mathsf{d}, H_{\text{eff}})$ of the form (67), given by Eqs. (78) and (80). The ansatz is of the general form (61), and uniqueness of the canonical form [4] identifies it as the canonical form of $\mathcal{L}_t$. Therefore:

Proposition 4.1. *At every time at which Λ_t is invertible, the canonical Hamiltonian of the Ising channel is diagonal in the configuration basis, and its decoherence matrix is supported on* $\operatorname{span}\{F_1, F_2, F_3\}$. *At most three canonical rates are nonzero; the remaining twelve vanish identically.*

No off-diagonal operator, and hence no operator that transfers population between configurations, appears in the generator. This is consistent with the fixed populations, $\Gamma_{aa}(t) = 1$.

4.6 Dissipative part: the coherence magnitudes

The six difference vectors and Eq. (74) give

pair ab	Δ_{ab}	$\kappa_{ab} = \frac{1}{2}\Delta^{\mathsf{T}}_{ab} S \Delta_{ab}$
$00, 01$	$(0, 1, 1)$	$\frac{1}{2}(S_{22} + S_{33} + 2S_{23})$
$10, 11$	$(0, 1, -1)$	$\frac{1}{2}(S_{22} + S_{33} - 2S_{23})$
$00, 10$	$(1, 0, 1)$	$\frac{1}{2}(S_{11} + S_{33} + 2S_{13})$
$01, 11$	$(1, 0, -1)$	$\frac{1}{2}(S_{11} + S_{33} - 2S_{13})$
$00, 11$	$(1, 1, 0)$	$\frac{1}{2}(S_{11} + S_{22} + 2S_{12})$
$01, 10$	$(1, -1, 0)$	$\frac{1}{2}(S_{11} + S_{22} - 2S_{12})$

This linear map from the six entries of S to the six κ_{ab} is invertible. Differences within each row pair give the off-diagonal entries; with the pair sums

$$\begin{aligned} K_A &= \kappa_{00,10} + \kappa_{01,11} = S_{11} + S_{33}, \qquad & K_B &= \kappa_{00,01} + \kappa_{10,11} = S_{22} + S_{33}, \\ K_{AB} &= \kappa_{00,11} + \kappa_{01,10} = S_{11} + S_{22}, & & \end{aligned} \tag{81}$$

for coherences in which only A, only B, or both probes flip, the inverse is

$$\begin{aligned} S_{11} &= \tfrac{1}{2}(K_A + K_{AB} - K_B), & S_{12} &= \tfrac{1}{2}(\kappa_{00,11} - \kappa_{01,10}), \\ S_{22} &= \tfrac{1}{2}(K_B + K_{AB} - K_A), & S_{13} &= \tfrac{1}{2}(\kappa_{00,10} - \kappa_{01,11}), \\ S_{33} &= \tfrac{1}{2}(K_A + K_B - K_{AB}), & S_{23} &= \tfrac{1}{2}(\kappa_{00,01} - \kappa_{10,11}). \end{aligned} \tag{82}$$

This is an explicit alternative to $S = \Re(G^{\mathsf{T}}\eta G)$. The entries have a direct reading:

- S_{11} and S_{22} describe dephasing of each probe along z.
- $S_{12} > 0$ when the coherence between 00 and 11 decays faster than that between 01 and 10, as for noise acting equally on both probes; $S_{12} < 0$ in the opposite, anticorrelated case.
- S_{13} (S_{23}) measures how the dephasing of A (B) depends on the configuration of the other probe.
- S_{33} is the diagonal decoherence-matrix coefficient associated with $F_3 = \sigma_A^z\sigma_B^z/2$.

These entries are coefficients in the chosen operator basis, rather than canonical rates. They coincide with canonical rates when the full matrix $\mathsf{d} = S + iA$ is diagonal in that basis (Section 4.8). If Γ factorizes into two independent single-probe dephasing maps, $\Gamma_{(s_A s_B),(s'_A s'_B)} = \alpha_{s_A s'_A}\beta_{s_B s'_B}$, then $\kappa_{00,10} = \kappa_{01,11} = \kappa^A$, $\kappa_{00,01} = \kappa_{10,11} = \kappa^B$ and $\kappa_{00,11} = \kappa_{01,10} = \kappa^A + \kappa^B$, so Eq. (82) gives $S = \mathrm{diag}(2\kappa^A, 2\kappa^B, 0)$. Nonzero S_{12}, S_{13}, S_{23} or S_{33} therefore signal dephasing correlated through the common bulk.

For $a \neq b$, each Δ_{ab} has squared norm 2. With the unit vector $\mathbf{e}_{ab} = \Delta_{ab}/\sqrt{2}$ and the normalized operator $F_{ab} = \sum_\mu (\mathbf{e}_{ab})_\mu F_\mu$,

$$\kappa_{ab} = \mathbf{e}_{ab}^{\mathsf{T}} S\, \mathbf{e}_{ab} = \mathbf{e}_{ab}^{\mathsf{T}} \mathsf{d}\, \mathbf{e}_{ab}, \tag{83}$$

since $\mathbf{e}^{\mathsf{T}} A \mathbf{e} = 0$. Each coherence decay rate is thus the decoherence matrix projected onto one fixed traceless diagonal operator; for example, $F_{01,10} = (\sigma_A^z - \sigma_B^z)/(2\sqrt{2})$ and $F_{00,11} = (\sigma_A^z + \sigma_B^z)/(2\sqrt{2})$.

4.7 Coherent part: the coherence phases

The energies h_a cancel from a sum of phase rates around a closed loop of three configurations. For an oriented triple (a, b, c), Eq. (75) gives

$$\mathcal{C}_{abc} \equiv \omega_{ab} + \omega_{bc} + \omega_{ca} = \mathbf{g}^{a\mathsf{T}} A \mathbf{g}^b + \mathbf{g}^{b\mathsf{T}} A \mathbf{g}^c + \mathbf{g}^{c\mathsf{T}} A \mathbf{g}^a, \tag{84}$$

because $(h_a - h_b) + (h_b - h_c) + (h_c - h_a) = 0$. Write A through a vector $\mathbf{a} = (a_1, a_2, a_3)$, $A_{\mu\nu} = \epsilon_{\mu\nu\lambda} a_\lambda$, so that $\mathbf{x}^{\mathsf{T}} A \mathbf{y} = \mathbf{a} \cdot (\mathbf{x} \times \mathbf{y})$. Since $(\mathbf{g}^b - \mathbf{g}^a) \times (\mathbf{g}^c - \mathbf{g}^a) = \mathbf{g}^a \times \mathbf{g}^b + \mathbf{g}^b \times \mathbf{g}^c + \mathbf{g}^c \times \mathbf{g}^a$,

$$\mathcal{C}_{abc} = \mathbf{a} \cdot \left[(\mathbf{g}^b - \mathbf{g}^a) \times (\mathbf{g}^c - \mathbf{g}^a) \right]. \tag{85}$$

The four configurations have three independent loops. With $\mathbf{g}^{01} - \mathbf{g}^{00} = (0, -1, -1)$, $\mathbf{g}^{10} - \mathbf{g}^{00} = (-1, 0, -1)$ and $\mathbf{g}^{11} - \mathbf{g}^{00} = (-1, -1, 0)$, the cross products are $(1, 1, -1)$, $(-1, 1, -1)$ and $(-1, 1, 1)$, respectively, for the loops

$$\begin{aligned} \mathcal{C}_1 &\equiv \mathcal{C}_{00,01,10} = a_1 + a_2 - a_3, \\ \mathcal{C}_2 &\equiv \mathcal{C}_{00,01,11} = -a_1 + a_2 - a_3, \\ \mathcal{C}_3 &\equiv \mathcal{C}_{00,10,11} = -a_1 + a_2 + a_3, \end{aligned} \tag{86}$$

which determine

$$a_1 = \tfrac{1}{2}(\mathcal{C}_1 - \mathcal{C}_2), \qquad a_2 = \tfrac{1}{2}(\mathcal{C}_1 + \mathcal{C}_3), \qquad a_3 = \tfrac{1}{2}(\mathcal{C}_3 - \mathcal{C}_2). \tag{87}$$

This agrees with $A = \Im(G^{\mathsf{T}}\eta G)$ and makes its physical content explicit: the matrix A carries the part of the phase evolution that cannot be absorbed into four configuration energies, and it vanishes iff all loop sums $\mathcal{C}_{abc}$ vanish. With A fixed, the remaining phase rates $\zeta_{ab} = -\omega_{ab} + \mathbf{g}^{a\mathsf{T}} A\, \mathbf{g}^{b}$ equal $h_a - h_b$ and reproduce Eq. (80), since $\frac{1}{4}\sum_b \zeta_{ab} = -\frac{1}{4}\sum_b \omega_{ab}$. The Hamiltonian is a configuration-dependent energy shift of the probes induced by the bulk. For a factorized Γ (previous subsection) every loop sum vanishes, because each phase rate is a sum of single-probe phase rates that cancel around a closed loop; hence $A = 0$ in that case.

4.8 Canonical rates and Lindblad operators

Diagonalize the Hermitian matrix of Eq. (78) at each regular time,

$$\mathsf{d}(t) = V(t)\,\mathrm{diag}(\gamma_1(t), \gamma_2(t), \gamma_3(t))V(t)^\dagger, \qquad V^\dagger V = I_3, \tag{88}$$

so that

$$\boxed{\{\gamma_k(t)\}_{k=1}^{3} = \mathrm{spec}\big[S(t) + iA(t)\big], \qquad \gamma_4, \ldots, \gamma_{15} = 0.} \tag{89}$$

We order these three rates as $\gamma_1 \le \gamma_2 \le \gamma_3$. The additional twelve zeros are listed separately, so the indices $1, \ldots, 15$ do not constitute a globally sorted list. Define

$$\boxed{L_k(t) = \sum_{\mu=1}^{3} V_{\mu k}(t) F_\mu = \frac{1}{2}\big[V_{1k}\sigma_A^z + V_{2k}\sigma_B^z + V_{3k}\sigma_A^z\sigma_B^z\big].} \tag{90}$$

Substituting $\mathsf{d}_{\mu\nu} = \sum_k V_{\mu k}\gamma_k V_{\nu k}^*$ into Eq. (67) and using $F_\nu^\dagger = F_\nu$,

$$\sum_{\mu\nu} \mathsf{d}_{\mu\nu} F_\mu X F_\nu = \sum_k \gamma_k \Big(\sum_\mu V_{\mu k} F_\mu\Big) X \Big(\sum_\nu V_{\nu k} F_\nu\Big)^\dagger = \sum_k \gamma_k L_k X L_k^\dagger, \tag{91}$$

and likewise $\sum_{\mu\nu} \mathsf{d}_{\mu\nu} F_\nu F_\mu = \sum_k \gamma_k L_k^\dagger L_k$. The L_k are traceless and satisfy $\mathrm{Tr}(L_j^\dagger L_k) = \sum_{\mu\nu} V_{\mu j}^* V_{\nu k}\delta_{\mu\nu} = \delta_{jk}$. They are diagonal in the configuration basis,

$$L_k(t)|a\rangle = \Big(\sum_\mu V_{\mu k}(t) g_\mu^a\Big)|a\rangle, \tag{92}$$

so each L_k imprints a configuration-dependent amplitude and phase. If $A = 0$, the eigenvectors can be chosen real and the L_k are Hermitian; for $A \neq 0$ they are diagonal but generally non-Hermitian. For nondegenerate rates each L_k is unique up to a phase; within a degenerate rate subspace, orthonormal eigenoperators may be mixed by a unitary transformation [4]. Ordering rates at each time does not track an eigenoperator continuously through a crossing.

The canonical master equation of the Ising channel is therefore

$$\dot\rho = -i[H_{\mathrm{eff}}(t), \rho] + \sum_{k=1}^{3} \gamma_k(t)\Big(L_k(t)\rho L_k^\dagger(t) - \tfrac{1}{2}\{L_k^\dagger(t)L_k(t), \rho\}\Big), \tag{93}$$

with H_{eff}, γ_k and L_k determined by $\Gamma(t)$ alone. All operators in it are diagonal and mutually commute, so it fixes the populations and is unital. The L_k are not raising or lowering operators: such operators would move population, which Proposition 4.1 excludes for this channel.

4.9 The generator at the initial time

By Eq. (34), $\eta_{ab}(0) = \dot\Gamma_{ab}(0) = -i(\varepsilon_a - \varepsilon_b)$. Hence $\kappa_{ab}(0) = 0$ for every pair, and the invertible map of Eq. (82) gives $S(0) = 0$. In matrix form, $\eta(0) = -i\varepsilon\mathbf{1}^{\mathsf{T}} + i\mathbf{1}\varepsilon^{\mathsf{T}}$ with $\varepsilon = (\varepsilon_a)_a$, so Eq. (78) and $\mathbf{1}^{\mathsf{T}}G = 0$ give

$$\mathsf{d}(0) = G^{\mathsf{T}}\eta(0)G = 0, \qquad h_a(0) = \varepsilon_a - \frac{1}{4}\sum_b \varepsilon_b. \tag{94}$$

All canonical rates vanish at $t = 0$, and the initial motion is purely coherent, generated by the bulk-averaged conditional energies $\mathrm{Tr}_E(\omega_E H_a)$. This is the general behaviour of a product initial state under a finite-dimensional unitary dilation; dissipative contributions to d appear only at $t > 0$.

4.10 Sign of the rates and CP divisibility

Following Hall *et al.* [4], a time-local master equation is called Markovian at time t iff all canonical rates are nonnegative, and non-Markovian iff at least one rate is strictly negative. On an interval where the map is differentiable and invertible, nonnegative rates throughout the interval are equivalent to CP divisibility [4]:

$$\text{CP-divisible} \iff \mathsf{d}(t) \succeq 0 \iff \gamma_k(t) \geq 0 \quad \forall k \text{ (almost everywhere)}. \tag{95}$$

A negative rate does not invalidate Eq. (93), and Λ_t remains CPTP for every $t \geq 0$, because $\Gamma(t) \succeq 0$ with unit diagonal; it signals that the intermediate maps near t are not completely positive, consistent with the finite-interval test of Eq. (48).

For the Ising channel the sign structure follows from S and A: write $\gamma_{\min} = \gamma_1$ and $\gamma_{\max} = \gamma_3$ for the extreme eigenvalues of the 3×3 diagonal-operator block.

1. The smallest rate obeys $\gamma_{\min} \leq \lambda_{\min}(S)$. Indeed $S = (\mathsf{d} + \mathsf{d}^{\mathsf{T}})/2$, and $\mathsf{d}^{\mathsf{T}} = \mathsf{d}^*$ has the same spectrum as d, so Weyl's inequality gives $\lambda_{\min}(S) \geq \frac{1}{2}[\lambda_{\min}(\mathsf{d}) + \lambda_{\min}(\mathsf{d}^{\mathsf{T}})] = \gamma_{\min}$. A non-positive-semidefinite S therefore implies a negative canonical rate; the phase part A cannot restore positivity.

2. By Eq. (83) and the Rayleigh-quotient bounds for a Hermitian matrix and a unit vector, for every distinct pair $a \neq b$,

$$\gamma_{\min}(t) \leq \kappa_{ab}(t) \leq \gamma_{\max}(t). \tag{96}$$

   Hence a negative rate occurs whenever any coherence magnitude $|\Gamma_{ab}|$ increases. The converse fails: a negative rate can occur with every κ_{ab} positive. Likewise $\gamma_{\min} \leq S_{\mu\mu} = \mathbf{u}_\mu^{\mathsf{T}} \mathsf{d} \mathbf{u}_\mu$ for each unit vector $\mathbf{u}_\mu$; in particular $K_{AB} > K_A + K_B$, i.e. $S_{33} < 0$, implies a negative rate even if every κ_{ab} is positive.

3. $S \succeq 0$ iff there is a real 3×3 matrix M with $S = M^{\mathsf{T}} M$, i.e. $\kappa_{ab} = \frac{1}{2}|M\Delta_{ab}|^2$ for all pairs. In that case, the six decay rates are one half of squared Euclidean distances between four points $M\mathbf{g}^a$ representing the configurations.

The complete condition is $\mathsf{d} = S + iA \succeq 0$. Although $S \succeq 0$ and nonnegative κ_{ab} are necessary for this condition, neither alone is sufficient. For example, $S = I_3$ with $A_{12} = -A_{21} = 2$ and all other entries of A zero has $\kappa_{ab} = 1$ for all $a \neq b$, but $\mathrm{spec}(S + iA) = \{1 - 2, 1, 1 + 2\} = \{-1, 1, 3\}$.

4.11 Singular times

The rates are built from the logarithmic derivatives $\eta_{ab} = \dot{\Gamma}_{ab}/\Gamma_{ab}$. The inverse-map construction fails when a coherence factor vanishes [4]. For the finite system with a time-independent Hamiltonian, each $\Gamma_{ab}(t)$ is analytic in t (a finite sum of exponentials). Near a zero of order $r \geq 1$ at t_*,

$$\Gamma_{ab}(t) = c(t - t_*)^r[1 + O(t - t_*)], \quad c \neq 0, \qquad \eta_{ab}(t) = \frac{r}{t - t_*} + O(1). \tag{97}$$

The second relation follows from $\eta_{ab} = \frac{d}{dt} \ln \Gamma_{ab} = r/(t - t_*) + \frac{d}{dt} \ln[1 + O(t - t_*)]$, where the last term is analytic. Since the pole is real, $\kappa_{ab} = -r/(t - t_*) + O(1)$ diverges, whereas

$\omega_{ab} = \Im\eta_{ab} = O(1)$ remains bounded. For odd r the phase $\arg\Gamma_{ab}$ jumps by π across t_*, because $(t-t_*)^r$ changes sign. By Eq. (96), at least one canonical rate becomes unbounded: $\gamma_{\max} \to +\infty$ as $t \to t_*^-$ and $\gamma_{\min} \to -\infty$ as $t \to t_*^+$. At t_* the rates defined by $\dot{\Lambda}_t\Lambda_t^{-1}$ are undefined. The time-local form applies on neighboring intervals where all coherence factors are nonzero. The state and its derivative remain bounded: $\rho_{ab}(t) = \Gamma_{ab}(t)r_{ab}$ and $|\dot{\rho}_{ab}| \le |\dot{\Gamma}_{ab}||r_{ab}|$. According to Eq. (60), κ_{ab} measures a fractional rather than an absolute change. Its signed integral is finite on a closed interval containing no zeros, as shown below Eq. (60). At an exact zero the corresponding one-sided logarithmic integrals diverge; a principal-value cancellation does not define a finite ordinary integral across the singularity. Lemma 3.1 shows that this divergence is unavoidable: an invertible map cannot reach a singular one under a locally integrable generator.

4.12 Computational procedure and consistency tests

For each chosen $(N, J, \Delta, g_A, g_B, \omega_E)$ and time t:

1. Build the four bulk matrices H_a of Eq. (19) and propagate $U_a(t)$, or propagate the vectors of a decomposition of ω_E.
2. Compute the six complex values $\Gamma_{ab} = \mathrm{Tr}(\omega_E U_b^\dagger U_a)$, $a < b$ [Eq. (26)].
3. Compute $\dot{\Gamma}_{ab} = -i\,\mathrm{Tr}[\omega_E U_b^\dagger (H_a - H_b) U_a]$ [Eq. (32)].
4. If each $\Gamma_{ab} \neq 0$, set $\eta_{ab} = \dot{\Gamma}_{ab}/\Gamma_{ab}$, $\eta_{ba} = \eta_{ab}^*$ and $\eta_{aa} = 0$.
5. Form $\mathsf{d} = G^{\mathsf{T}}\eta G$ and $h_a = -\frac{1}{4}\sum_b \Im\eta_{ab}$ [Eqs. (78) and (80)].
6. Diagonalize only the 3×3 Hermitian matrix d, obtaining γ_k and the diagonal operators L_k [Eq. (90)].

No 16×16 diagonalization is required. At every regular time the calculation can be checked independently:

1. $\Gamma = \Gamma^\dagger \succeq 0$, $\Gamma_{aa} = 1$; $\eta = \eta^\dagger$, $\eta_{aa} = 0$.
2. $\mathsf{d} = \mathsf{d}^\dagger$; $\sum_a h_a = 0$; $\mathrm{Tr}(L_j^\dagger L_k) = \delta_{jk}$.
3. Reconstruct all six complex multipliers from the canonical form, η_{ab}^{rec} given by the right side of Eq. (71), and check $\eta_{ab}^{\mathrm{rec}} = \dot{\Gamma}_{ab}/\Gamma_{ab}$.
4. Apply $\mathcal{L}_t$ to all sixteen E_{ab} and check $\mathcal{L}_t(E_{ab}) = \eta_{ab}E_{ab}$.

These tests verify the *entire two-probe channel generator*, not merely its action on the special input $|{+}{+}\rangle$. Step 3 can also be replaced by a finite difference of Γ; the analytic form avoids the associated discretization error.

Part II
Exchange coupling

5 Exchange coupling: exact symmetry reduction

5.1 Total charge and Hamiltonian sectors

For the field-free model define

$$M_S = S_A^z + S_B^z, \qquad M_E = \sum_{j=1}^{N} S_j^z, \qquad M = M_S + M_E. \tag{98}$$

The bulk XXZ terms conserve M_E, and each exchange term $S_A^+ S_1^- + S_A^- S_1^+$ (or its right-contact analogue) transfers one unit between a probe and the bulk while conserving their sum. Hence $[H^{(\mathrm{ex})}, M] = 0$, while individual probe S^z values generally are not conserved. The total-charge sector m contains the basis states with $(N+2)/2 - m$ down spins, so

$$\dim \mathcal{H}_m = \binom{N+2}{(N+2)/2 - m}. \tag{99}$$

This Hamiltonian decomposition requires no assumption on the input; coherences between occupied sectors must still be retained. There is no four-branch controlled propagator of the form (20) for exchange coupling.

5.2 Bulk assumption and proof of phase covariance

For the channel reduction assume

$$[\omega_E, M_E] = 0. \tag{100}$$

Support in a single well-defined bulk magnetization sector is sufficient. An incoherent mixture of sectors also suffices. A coherent superposition of different bulk magnetizations generally does not.

Let $V_\theta = e^{-i\theta M_S}$ and $W_\theta = e^{-i\theta M_E}$, so that $V_\theta \otimes W_\theta = e^{-i\theta M}$. Conservation gives $[U_t, V_\theta \otimes W_\theta] = 0$, while Eq. (100) gives $W_\theta \omega_E W_\theta^\dagger = \omega_E$. Then

$$\begin{aligned}
\Lambda_t(V_\theta X V_\theta^\dagger) &= \mathrm{Tr}_E\big[U_t(V_\theta \otimes W_\theta)(X \otimes \omega_E)(V_\theta \otimes W_\theta)^\dagger U_t^\dagger\big] \\
&= \mathrm{Tr}_E\big[(V_\theta \otimes W_\theta) U_t (X \otimes \omega_E) U_t^\dagger (V_\theta \otimes W_\theta)^\dagger\big] \\
&= V_\theta \ \mathrm{Tr}_E\big[(I \otimes W_\theta) U_t (X \otimes \omega_E) U_t^\dagger (I \otimes W_\theta^\dagger)\big] \ V_\theta^\dagger = V_\theta \Lambda_t(X) V_\theta^\dagger,
\end{aligned} \tag{101}$$

where the last step uses cyclicity of the partial trace for operators acting only on E. Hence

$$\boxed{\Lambda_t(V_\theta X V_\theta^\dagger) = V_\theta \Lambda_t(X) V_\theta^\dagger.} \tag{102}$$

This is $U(1)$ phase covariance. It permits exchange of magnetization with the bulk; it does not imply $\Lambda_t^*(M_S) = M_S$ or unitality.

5.3 Five superoperator blocks

The probe charges are $(m_{00}, m_{01}, m_{10}, m_{11}) = (1, 0, 0, -1)$. The operator E_{ab} has coherence order $q = m_a - m_b$, meaning $[M_S, E_{ab}] = qE_{ab}$ and $V_\theta E_{ab} V_\theta^\dagger = e^{-iq\theta} E_{ab}$. Write $\Lambda_t(E_{ab}) = \sum_{c,d} c_{cd} E_{cd}$.

Covariance gives

$$\sum_{c,d} e^{-i(m_c-m_d)\theta} c_{cd} E_{cd} = V_\theta \Lambda_t(E_{ab}) V_\theta^\dagger = \Lambda_t(V_\theta E_{ab} V_\theta^\dagger) = e^{-iq\theta} \sum_{c,d} c_{cd} E_{cd} \tag{103}$$

for every θ. Hence $c_{cd} = 0$ unless $m_c - m_d = q$, and

$$\Lambda_t(\mathcal{V}_q) \subseteq \mathcal{V}_q, \qquad \mathcal{V}_q = \mathrm{span}\{E_{ab} : m_a - m_b = q\}. \tag{104}$$

The ordered bases $\mathcal{B}_q$ of these spaces are

q	Dimension	Basis of $\mathcal{V}_q$
0	6	$E_{00,00}, E_{01,01}, E_{10,10}, E_{11,11}, E_{01,10}, E_{10,01}$
$+1$	4	$E_{00,01}, E_{00,10}, E_{01,11}, E_{10,11}$
-1	4	Adjoints of the $q = +1$ basis
$+2$	1	$E_{00,11}$
-2	1	$E_{11,00}$

Thus the 16×16 superoperator has the exact reduction

$$\boxed{\Lambda_t \simeq T^{(-2)} \oplus T^{(-1)} \oplus T^{(0)} \oplus T^{(+1)} \oplus T^{(+2)}, \quad (d_q) = (1, 4, 6, 4, 1).} \tag{105}$$

Operators of the same order may mix. Hermiticity preservation, $\Lambda_t(X^\dagger) = \Lambda_t(X)^\dagger$, relates the opposite orders, since $E_{ab}^\dagger = E_{ba}$ has order $-q$. Using Eq. (9), $(K_{r\nu} E_{ab} K_{r\nu}^\dagger)_{cd} = (K_{r\nu})_{ca} (K_{r\nu})^*_{db}$, so in the ordered basis $\mathcal{B}_q$ the exact entries are

$$T^{(q)}_{cd,ab}(t) = \sum_{r,\nu} (K_{r\nu})_{ca} (K_{r\nu})^*_{db}, \qquad E_{ab}, E_{cd} \in \mathcal{B}_q. \tag{106}$$

Equation (106) is a formal construction, not a closed evaluation of these functions for general interacting XXZ dynamics.

5.4 Choi blocks

By Eq. (12), $J_t = \frac{1}{4} \sum_{a,b} \sum_{c,d} T_{cd,ab} E_{cd} \otimes E_{ab}$, with nonzero coefficients only for $m_c - m_d = m_a - m_b$. Since $M_S^\mathsf{T} = M_S$ is diagonal, $[M_S \otimes I_R - I_{AB} \otimes M_S^\mathsf{T}, E_{cd} \otimes E_{ab}] = [(m_c - m_d) - (m_a - m_b)] E_{cd} \otimes E_{ab}$, which vanishes on every nonzero term. Hence the normalized Choi state satisfies

$$[J_t, M_S \otimes I_R - I_{AB} \otimes M_S^\mathsf{T}] = 0. \tag{107}$$

Group output–reference kets $|c\rangle|a\rangle$ by $m_c - m_a$; the term $E_{cd} \otimes E_{ab} = |ca\rangle\langle db|$ connects $m_c - m_a$ to $m_d - m_b$, which are equal. The numbers of pairs (c, a) with $m_c - m_a = k$ are $1, 4, 6, 4, 1$ for $k = -2, \ldots, 2$, so the Choi blocks have the same sizes. Since J_t is Hermitian, the blocks contain at most $1 + 16 + 36 + 16 + 1 = 70$ real parameters before trace preservation and positivity are imposed, rather than 256. Choi blocks and superoperator blocks are different rearrangements of channel coefficients and should not be identified as matrices or assigned the same spectra.

5.5 Kraus operators with definite charge transfer

Under Eq. (100), diagonalize ω_E within its charge sectors: $M_E|\phi_\nu\rangle = \mu_\nu|\phi_\nu\rangle$. Choose $M_E|r\rangle = \mu_r|r\rangle$. From $[U_t, M] = 0$,

$$\begin{aligned} M_S K_{r\nu} &= \sqrt{p_\nu}\, {}_E\langle r|(M - M_E) U_t|\phi_\nu\rangle_E = \sqrt{p_\nu}\, {}_E\langle r|U_t M|\phi_\nu\rangle_E - \mu_r K_{r\nu} \\ &= \sqrt{p_\nu}\, {}_E\langle r|U_t(M_S + \mu_\nu)|\phi_\nu\rangle_E - \mu_r K_{r\nu} = K_{r\nu} M_S + (\mu_\nu - \mu_r) K_{r\nu}. \end{aligned} \tag{108}$$

Hence

$$[M_S, K_{r\nu}] = (\mu_\nu - \mu_r)K_{r\nu}, \qquad (K_{r\nu})_{ca} \neq 0 \implies m_c - m_a = \mu_\nu - \mu_r. \tag{109}$$

Since $m_c - m_a \in \{-2, \ldots, 2\}$, only transfers between -2 and $+2$ can act on the two probes; Kraus operators with other transfers vanish. Under V_θ, each $K_{r\nu}$ acquires the scalar phase $e^{-i\theta(\mu_\nu - \mu_r)}$, which cancels in $K_{r\nu} X K_{r\nu}^\dagger$; this reproduces Eq. (102). The transfer label belongs to a Kraus operator; coherence order labels an input or output matrix unit. Their roles in the Choi and superoperator reductions are related but distinct.

5.6 Reduced states at fixed total magnetization

If the *global* initial state is supported in one total-M sector, the reduced probe state has the exact form

$$\rho_{AB}(t) = \begin{pmatrix} p_{00} & 0 & 0 & 0 \\ 0 & p_{01} & c & 0 \\ 0 & c^* & p_{10} & 0 \\ 0 & 0 & 0 & p_{11} \end{pmatrix}. \tag{110}$$

For a pure global ket $|\Psi(t)\rangle$ with $M|\Psi\rangle = m|\Psi\rangle$ (preserved by $[U_t, M] = 0$), resolve its four probe components, $|\Psi(t)\rangle = \sum_a |a\rangle \otimes |\psi_a(t)\rangle$. Since $M_S|a\rangle = m_a|a\rangle$, each bulk ket satisfies $M_E|\psi_a\rangle = (m - m_a)|\psi_a\rangle$; the charges are $m-1, m, m, m+1$. The reduced state has entries $\rho_{ab} = \langle\psi_b|\psi_a\rangle$, and bulk kets of different charge are orthogonal. This eliminates every coherence except $01 \leftrightarrow 10$, with $c = \langle\psi_{10}|\psi_{01}\rangle$. Linearity extends the conclusion to mixtures within the sector. The populations and c remain time dependent. For this X state with $\rho_{00,11} = 0$, the Wootters formula reduces to

$$C = 2\max\{0, |c| - \sqrt{p_{00}p_{11}}\}. \tag{111}$$

This restriction does *not* apply to the texture protocol merely because the bulk has fixed magnetization. The input $|{+}{+}\rangle$ contains probe charges $+1, 0, -1$ with probabilities $1/4, 1/2, 1/4$, and hence occupies three total-charge sectors for a fixed-charge bulk. Their coherences cannot be discarded. In particular, f_1 itself is not of the form (110).

Reflection and global spin flip give additional covariance only when both the Hamiltonian and bulk preparation have the required symmetry. Equal contacts or a symmetric Hamiltonian alone do not impose that symmetry on an arbitrary reduced trajectory.

5.7 Exchange texture and the limits of one input

Equations (15), (16) and (17) hold for exchange coupling as well. However, ρ_f is generally not a Schur multiplier matrix and does not determine the exchange channel. Under Eq. (100), $f_1 = \frac{1}{4}\sum_{a,b} E_{ab}$ has a nonzero component in every $\mathcal{V}_q$, but one output determines only each block's action on that particular component. It cannot determine a general 6×6 or 4×4 block.

There is one exact scalar exception. Since $\mathcal{V}_{+2}$ is one-dimensional, write $\Lambda_t(E_{00,11}) = \xi_{+2}(t)E_{00,11}$. By the block selection rule, the $E_{00,11}$ component of $\rho_f = \frac{1}{4}\sum_{a,b}\Lambda_t(E_{ab})$ comes only from $\frac{1}{4}\Lambda_t(E_{00,11})$. Then

$$\xi_{+2}(t) = 4[\rho_f(t)]_{00,11}, \qquad \xi_{-2}(t) = \xi_{+2}(t)^*, \tag{112}$$

the second relation by Hermiticity preservation. This inference requires covariance; it is not guaranteed for a bulk coherent across magnetization sectors or for transverse-field dynamics. The equality $F_e = p_f$ is likewise not a general exchange-channel identity, although $F_{\text{avg}} = (4F_e + 1)/5$ still holds.

5.8 Applicability of the symmetry reduction

The exchange block decomposition needs both $[H, M_S + M_E] = 0$ and $[\omega_E, M_E] = 0$. Transverse bulk fields generally violate total-magnetization conservation and couple different total-magnetization sectors. Coherence between bulk magnetization sectors generally removes channel phase covariance, even for a conserving Hamiltonian. The Ising Schur channel and $\Gamma = 4\rho_f$ remain valid in both cases, because the probe z components are still conserved.

5.9 Summary of the exchange channel structure

For exchange coupling, conservation of total magnetization and a magnetization-invariant bulk preparation give five coherence-order blocks. Symmetry fixes their dimensions and forbids mixing between different orders, but leaves the entries within each block to be calculated. The output from $|{+}{+}\rangle$ determines the texture and the one-dimensional highest-order response; it does not specify the full exchange channel. A general closed expression for the interacting exchange coefficients remains unavailable.

A State diagnostics and their interpretation

For $\rho = \rho_{AB}$, the state purity and entropy are

$$P = \operatorname{Tr} \rho^2, \qquad S(\rho) = -\sum_k \lambda_k \ln \lambda_k, \qquad 0 \ln 0 = 0. \tag{113}$$

Logarithms are natural. The probe mutual information is $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho)$, with $\rho_A = \operatorname{Tr}_B \rho$ and $\rho_B = \operatorname{Tr}_A \rho$. It includes classical and quantum correlations.

Probe negativity and concurrence are

$$\mathcal{N}_{A|B}(\rho) = \frac{\|\rho^{T_B}\|_1 - 1}{2}, \tag{114}$$

$$\widetilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \qquad C(\rho) = \max(0, r_1 - r_2 - r_3 - r_4), \tag{115}$$

where the r_i are the decreasing nonnegative square roots of the eigenvalues of $\rho\widetilde{\rho}$ [5]. For two qubits, $C > 0$ and $\mathcal{N} > 0$ are equivalent tests of entanglement. They concern $A|B$, while $S(\rho_{AB})$ concerns the pair's marginal mixedness. Only for a globally pure state does $S(\rho_{AB})$ equal entanglement entropy across $AB|E$.

For the Ising texture input, Eq. (41) with $\sum_{a,b} |\Gamma_{ab}|^2 = 4 + 2\sum_{a<b} |\Gamma_{ab}|^2$ gives

$$P_f(t) = \frac{1}{4} + \frac{1}{8} \sum_{a<b} |\Gamma_{ab}(t)|^2, \qquad \frac{1}{4} \le P_f \le 1. \tag{116}$$

Its local probe coherences, obtained by tracing out the other probe, are

$$(\rho_A)_{01} = \frac{\Gamma_{00,10} + \Gamma_{01,11}}{4}, \qquad (\rho_B)_{01} = \frac{\Gamma_{00,01} + \Gamma_{10,11}}{4}, \tag{117}$$

and each local diagonal is $(1/2, 1/2)$. For a pure bulk, the global state is $\frac{1}{2}\sum_a |a\rangle \otimes U_a|\phi_0\rangle$; it is a product across $AB|E$ iff its four conditional bulk kets are proportional. For a mixed bulk, reduced entropy or purity alone does not quantify entanglement across that cut.

References

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